Tuesday, Nov 10 · Bell work and side boards · IEP

Rule card: Negative × negative = positive. When you divide, divide EVERY term.

Bell work

1. −3 − (−8)

2. y = −2x + 5. Find y when x = −3.

Level 1 · Integers

Example: −9 − (−4) → −9 + 4 = −5

💡 Minus a negative = plus.

a. −6 + 10

b. 4 − 9

c. −7 − (−3)

Level 2 · Distribute

Example: −3(x − 2) → −3·x and −3·(−2) → −3x + 6

💡 Draw an arrow from the outside number to EACH term.

a. 3(x − 4)

b. −2(x + 5)

c. −4(x − 3)

Level 3 · Divide every term

Example: $(-12x + 6) \div (-6) \rightarrow -12x \div -6 = 2x$ and $6 \div -6 = -1 \rightarrow 2x - 1$

💡 Split it: one division for each term.

a. $(6x + 12) \div 3$

b. $(-10x + 20) \div (-5)$

c. $(-6x - 9) \div (-3)$

Wednesday, Nov 11 · Bell work and side boards · IEP

Rule card: Undo backwards: + or − first, then × or ÷. Fraction in front of x → multiply by its flip.

Bell work

1. 8x − 4y = 20. Get y alone.

2. 5 − (−2)

Level 1 · Two-step equations

Example: 4x + 3 = 19 → subtract 3: 4x = 16 → divide by 4: x = 4

💡 Undo + or − first, then × or ÷.

a. 2x + 5 = 17

b. 3x − 7 = 14

c. x/4 + 2 = 5

Level 2 · Fractions: multiply by the flip

Example: (3/5)x = 6 → x = 6 · 5/3 = 10

💡 Flip the fraction and multiply.

a. (2/3)x = 8

b. (3/4)x = −9

c. −(1/2)x = 6

Level 3 · Find b

Example: $y = 4x + b$ through $(1, 9) \rightarrow 9 = 4(1) + b \rightarrow 9 = 4 + b \rightarrow b = 5$

💡 Put in x and y, then solve for b.

a. $y = 2x + b$ goes through $(3, 10)$

b. $y = -3x + b$ goes through $(2, 1)$

c. $y = \frac{1}{2}x + b$ goes through $(-4, 5)$

Thursday, Nov 12 · Bell work and side boards · IEP

Rule card: On a number line, left is less. Open ○ for < and >. Closed ● for ≤ and ≥.

Bell work

1. V = ℓwh. Solve for h.

2. Which is greater, −7 or −2?

Level 1 · < or >

Example: −6 __ −2 → −6 is farther left → −6 < −2

💡 Farther left = smaller.

a. −3 __ 2

b. −8 __ −5

c. −1 __ −10

Level 2 · Words to symbols

Example: at most 8 → x ≤ 8

💡 at least ≥ · at most ≤ · more than > · less than <

a. at least 5

b. no more than 12

c. more than −2

Level 3 · Is it a solution?

Example: $x - 2 > 3$ · try $x = 6$ → $6 - 2 = 4$, and $4 > 3$ → Yes

💡 Put the number in. True → Yes. False → No.

a. $x + 4 > 1$ · try $x = -3$

b. $x + 4 > 1$ · try $x = 0$

c. $2x - 1 \leq 5$ · try $x = 3$

Monday, Nov 16 · Bell work and side boards · IEP

Rule card: Same signs → positive. Different signs → negative. Multiply or divide by a negative → flip the sign.

Bell work

1. $3.75b + 25 \geq 400$

2. $(-6)(-2)$ and $-15 \div (-3)$

Level 1 · Signs

Example: $(-6)(-2) \rightarrow$ same signs $\rightarrow 12$

💡 Same signs +, different signs −.

a. $-4 \cdot 3$

b. $-20 \div (-4)$

c. $18 \div (-6)$

Level 2 · One step with negatives

Example: $-4x = 20 \rightarrow$ divide by $-4 \rightarrow x = -5$

💡 Divide by the number in front of x, sign and all.

a. $-3x = 12$

b. $-5x = -35$

c. $x/(-2) = 6$

Level 3 · Does the sign flip?

Example: $1 < 4$ · multiply by -3 → -3 and -12 → $-3 > -12$ → flips

💡 Do the math, then picture both on a number line.

a. $3 < 7$ · multiply both by -2

b. $-4 < 8$ · divide both by -4

c. $10 > 5$ · multiply both by -1

Tuesday, Nov 17 · Bell work and side boards · IEP

Rule card: slope = $(y_2 - y_1) \div (x_2 - x_1)$. Parallel: same slope. Perpendicular: flip it and change the sign.

Bell work

1. $-2(x - 3)$

2. $5 - 3x > 11$

Level 1 · Slope

Example: (1, 3) and (4, 9) → $(9 - 3) \div (4 - 1) = 6 \div 3 = 2$

💡 Label x_1, y_1, x_2, y_2 first.

a. (1, 2) and (3, 8)

b. (−2, 5) and (2, −3)

c. (−3, 1) and (3, −3)

Level 2 · Parallel and perpendicular slopes

Example: perpendicular to $y = 3x$ → $-1/3$

💡 Flip 3/1 to 1/3, then change the sign.

a. parallel to $y = 4x - 1$

b. perpendicular to $y = 2x + 3$

c. perpendicular to $y = -(3/5)x + 1$

Level 3 · Substitute, then plot

Example: g(x) = 3x at x = 2 → 3 · 2 = 6 → point (2, 6)

💡 Replace x with the number.

a. g(x) = 2x for x = −1, 0, 1, 2

b. h(x) = −x + 3 for x = −2, 0, 3

c. Plot both. Which is steeper?